

1 Introduction

In guitar, tone control is a major technical skill. Each musical instrument's quality of sound —its timbre—is differentiated by the varying composition of harmonic frequencies f_n which are present in addition to the fundamental frequency f_1 . In addition to the fundamental variation between instruments, the timbre also varies based on how notes are played (Perov et al., 2016). It is qualitatively observed that plucking near the bridge will create a harsher, “tinny” sound while plucking toward the bridge will create a fuller sound.

This investigation will explore how the quality of sound ^{as measured by} may change depending on where the string is plucked. The research question I have decided to investigate is: *How does changing the position of the pluck along a guitar scale affect the frequency of the harmonic with the highest intensity when a string with constant tension is plucked?* ^{could be made clearer by adding some more details}

1.1 Background

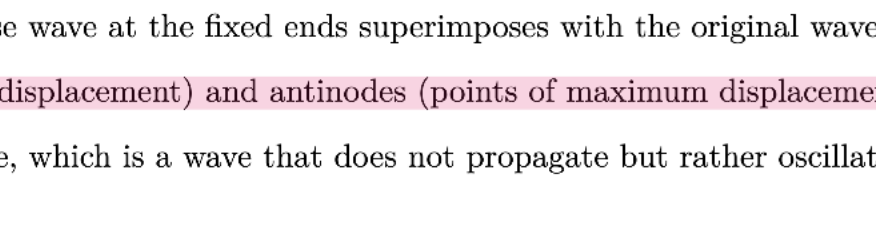


Figure 1: A guitar string of length L pulled a distance A at position x_1 (Perov et al., 2016)

A guitar string is a string with two fixed ends. When the string is plucked (as in Figure 1), a transverse wave is initiated that propagates along the length of the string. The reflection of the transverse wave at the fixed ends superimposes with the original wave, ^{forming nodes} (points of zero displacement) and antinodes (points of maximum displacement). This creates a standing wave, which is a wave that does not propagate but rather oscillates in place (Tsokos, 2014).

The first harmonic is a standing wave with a single loop made of two nodes and one antinode, which results in a sound wave at the fundamental frequency, given by the equation $f_1 = \frac{v}{2L}$, ^{where v is ? , f is ?, and L is ?}

The linear principle of superposition states that the net displacement of a system subjected to multiple independent disturbances (waves, in this case) is the algebraic sum of the displacements due to each disturbance. If $y_1(x, t)$ and $y_2(x, t)$ are two waveforms propagating along the same string, then the net waveform $y(x, t)$ ^{is given by the equation} $y(x, t) = y_1(x, t) + y_2(x, t)$ ^{A sketch/image will provide a better understanding of this.}

$$y(x, t) = y_1(x, t) + y_2(x, t)$$
 (1)

Due to this principle, multiple, independent standing waves are formed on the guitar string, which results in higher-order harmonics. Each subsequent harmonic is an integer multiple of the fundamental frequency (known as frequency quantization), given by

$$f_n = n f_1$$
 (2)

These individual waveforms superimpose linearly to form a composite waveform which is the sum of all individual harmonic waveforms. As a result, the sound emitted by a guitar string is a mixture of multiple harmonic frequencies.

To investigate how the picking length affects ^{the harmonic with the highest amplitude}, we must determine the variation of ^{the amplitude of each harmonic frequency} separately. The amplitude b_n of the n th harmonic ^{is} given by the equation

What does each variable stand for? $b_n = \frac{2A \sin(n\pi \frac{x_1}{L})}{\pi (1 - \frac{x_1^2}{L^2}) + n^2}$ *How would this help answer the relationship in the question?* (3)

where A is the distance the string is pulled when plucked and L is the string length (Perov et al., 2016).

By calculating b_n (Equation 3) for $n = 2 \dots 10$ at each variation of x_1 , and choosing the value of n that maximizes b_n , the non-fundamental harmonic frequency with the highest amplitude can be calculated using Equation 2.

Plotting the picking length x_1 against the non-fundamental harmonic frequency with the highest amplitude f results in a Gaussian-like function where the 2nd harmonic has the highest amplitude across picking lengths, except near the middle of the fretboard where the 3rd harmonic has the highest amplitude. ^{Once again, more visuals please}

Therefore, it is hypothesized that the non-fundamental harmonic frequency with the highest amplitude will remain constant across picking lengths x_1 , except with a peak around the middle of the fretboard ($x_1 = \frac{L}{2}$).

sketch of standing waves on guitar string

show distribution for each harmonic from perov paper

2 Methodology

2.1 Variables

Type	Variable	Specification
Independent Variable	Picking length x_1 (m), the distance in meters from the nut where the string is disturbed	$x_1 = \{0.01\text{m}, 0.08\text{m}, 0.15\text{m}, 0.22\text{m}, 0.29\text{m}, 0.36\text{m}, 0.43\text{m}, 0.50\text{m}, 0.57\text{m}, 0.64\text{m}\}$, varied by plucking at the picking length as indicated by a tape measure
	Frequency f of the non-fundamental harmonic with the highest amplitude (Hz)	Measured by recording the frequency f_n of the n th harmonic that has the highest amplitude through FFT (Fast Fourier Transform) analysis, where $n = 2 \dots 10$
Controlled Variables		
Control	Reason	Specification
Diameter, tension, age, and other material properties of the string	To minimize random error from the frequency response imparted by the pickups and guitar body.	The same high E string tuned to the standard $E = 329.63$ Hz will be used for all the experiments. By extension, the density will be kept constant by performing data collection in the same air-conditioned room (set to 23 °C)
	Different material properties of the string may change its harmonic composition.	
Guitar body and electronics	To minimize random error from the frequency response imparted by the pickups and guitar body.	The neck pickup on the same guitar will be used across trials.
Plectrum choice and picking technique	Different plectrums and picking techniques may affect the harmonic composition. This also controls A in Equation 3.	The same pick will be used across trials. The string will be plucked at a 45° angle to the fretboard, with the tip of the pick contacting the fretboard.

Table 1: Variables

explain what the equation models and preempt how it will be used what is the conceptual understanding behind this? then link back to this explanation in scientific reasoning to say that yeah, this is why only one picking length had this frequency.

equal spacing along string length

where to add this? It would be nice to include more conceptual description about how the non-fundamental harmonic with the highest amplitude helps improve the quality of sound.

2.2 Apparatus & Experimental Setup

- Electric guitar (6-string *Kiesel Guitars* “Osisis” model used)
- 1 set ^{change to be only high e string} electric guitar strings (*D’Addario XS Nickel XSE942 Super Light Coated* *Electric Guitar Strings* model used)
- 1 audio interface with instrument input (*Audient EVO4* used)
- 1 instrument cable
- 1 Computer with audio analysis and spreadsheet software (*MATLAB* and *Google Sheets* used)
- 1 tape measure
- 1 clamp stand
- Masking tape

Figure 2 presents the experimental setup used to collect data. The guitar is clamped down for ease of plucking, and the tape measure is placed directly on the 2nd and 3rd strings to be on the same axis as the 1st string (which is plucked).

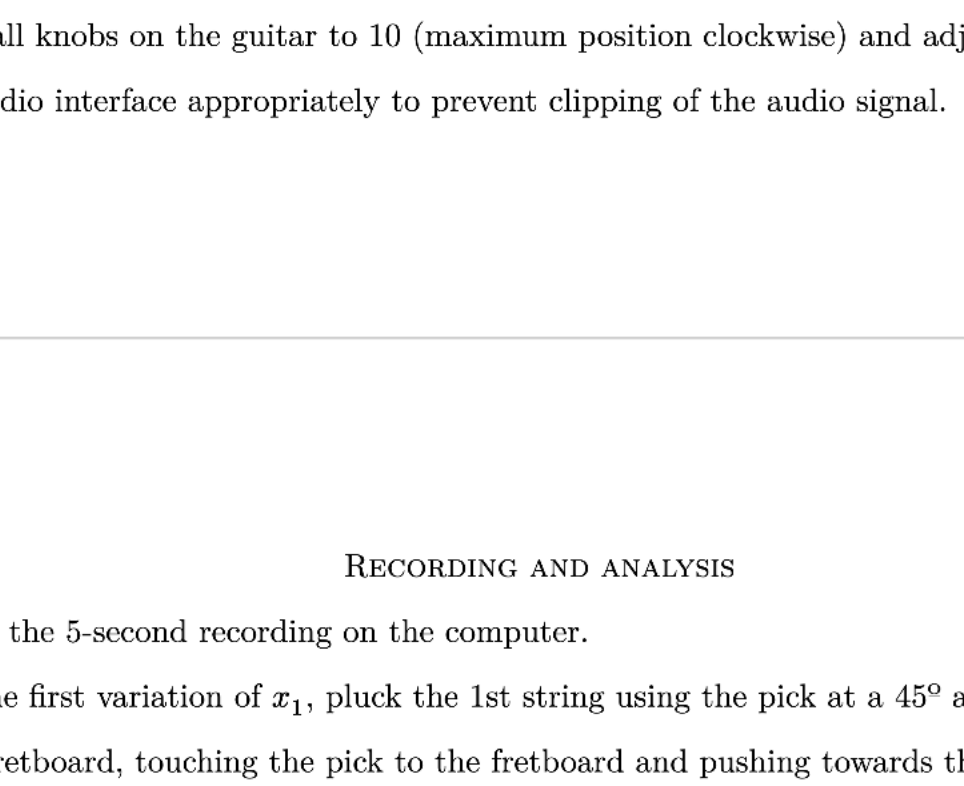


Figure 2: Labeled photograph of experimental setup

2.3 Experimental Procedure

PREPARATION

- Clamp the guitar’s head using the clamp stand to hold it in place.
- Lay the tape measure out along the fretboard on top of the 2nd and 3rd strings, securing it in place with the clamp stand and masking tape.
- Connect the guitar to the audio interface using the instrument cable, and connect the audio interface to the computer.
- Turn all knobs on the guitar to 10 (maximum position clockwise) and adjust the gain on the audio interface appropriately to prevent clipping of the audio signal.

RECORDING AND ANALYSIS

- Start the 5-second recording on the computer.
- At the first variation of x_1 , pluck the 1st string using the pick at a 45° angle to the fretboard, touching the pick to the fretboard and pushing towards the string perpendicularly.
- Wait for the recording period to end and ensure the data is saved.
- Using the frequency analysis software, ^{convert the time-domain data into frequency domain data.} ^{This is essential to any computer data collection need to be embedded in the method for clarity. just in the list? or as an additional section. can i annotate?}
- Identify ^{the} true fundamental frequency by searching for a peak in the data within 5% of the expected fundamental frequency $f_1 = 329.63$ Hz
- Calculate the frequency of each non-fundamental harmonic n for $n = 2 \dots 10$
- Calculate the absolute amplitude $|b_n|$ for the frequency at each harmonic, recording the frequency f_n that maximizes the absolute amplitude $|b_n|$.
- Repeat steps 5–11 four more times to complete five trials for the variation.
- Repeat steps 5–12 for the remaining variations, modifying the picking length x_1 according to Table 1.

Note: In data collection, steps 8–11 were automated using a custom MATLAB program. The program applies a Hann window to the recorded audio data to reduce noise interference and computes the Fourier transform of the windowed data using Welch’s method. The fundamental is identified by looking for peaks around the expected frequency, allowing for variations in tuning (MathWorks, 2023).

2.4 Safety, Ethical, and Environmental Considerations

This is a relatively safe experiment. As the guitar is directly connected to the audio interface, any voltages are low and are not likely to pose a safety risk. There is no risk of water damage to any electronics. New strings are not required for the experiment and no consumable materials are used, limiting this experiment’s environmental impacts.

3 Analysis

3.1 Raw Data - Time-Domain

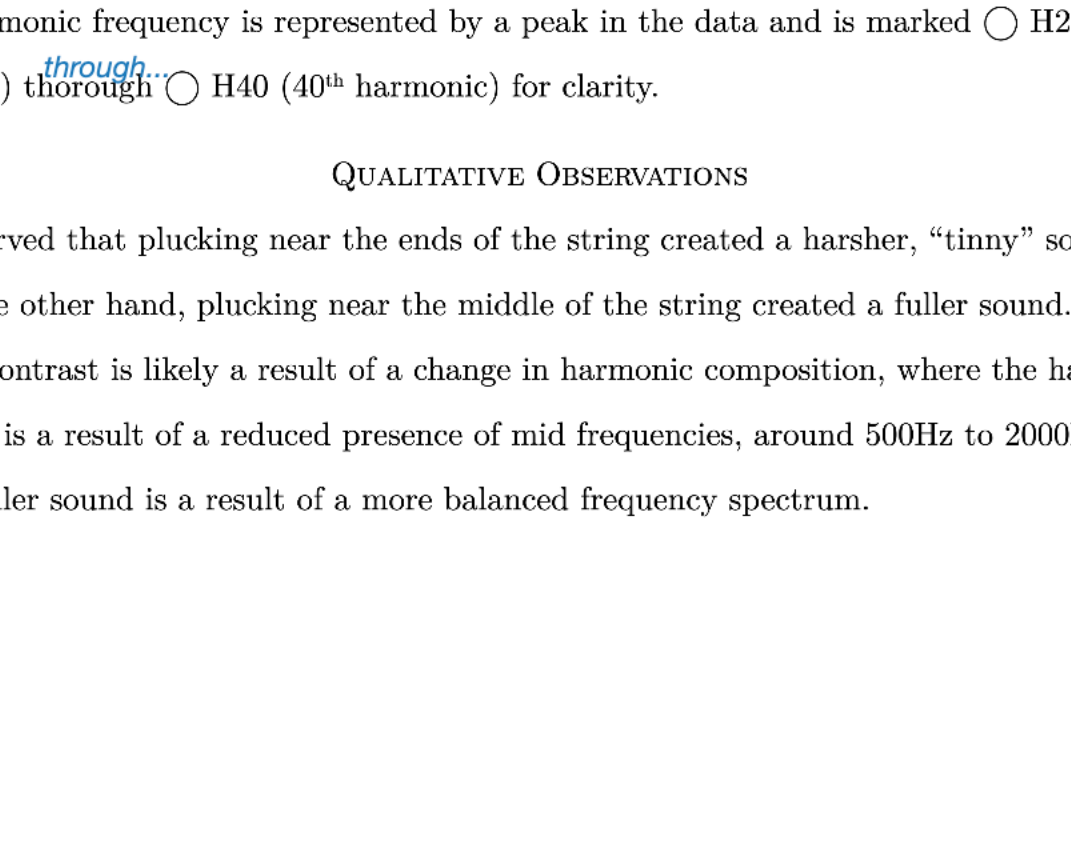


Figure 3: Example of frequency-domain data showing harmonic peaks

As an example, Figure 3 shows the frequency-domain data for a specific trial and variation, which has been converted by the program from the time series data (omitted because it cannot be represented meaningfully in text/table form).

Each harmonic frequency is represented by a peak in the data and is marked $\bigcirc$ H2 (2nd harmonic) ^{through} $\bigcirc$ H40 (40th harmonic) for clarity.

QUALITATIVE OBSERVATIONS

- I observed that plucking near the ends of the string created a harsher, “tinny” sound.
- On the other hand, plucking near the middle of the string created a fuller sound.
- This contrast is likely a result of a change in harmonic composition, where the harsher sound is a result of a reduced presence of mid frequencies, around 500Hz to 2000Hz, and the fuller sound is a result of a more balanced frequency spectrum.

3.2 Raw Data - Frequency-Domain

Plucking length x_1 (m)	Frequency with highest amplitude $f / \Delta f$ (Hz)				
	$\Delta x_1 = \pm 0.005\text{m}$	Trial 1	Trial 2	Trial 3	Trial 4
0.01		655.70	655.88	655.88	655.88
0.08		655.70	655.88	655.88	655.88
0.15		655.88	655.88	655.88	656.07
0.22		656.07	655.88	655.88	656.07
0.29		984.01	983.83	984.01	984.01
0.36		656.07	984.01 ¹	655.88	655.88
0.43		655.88	656.07	655.88	655.88
0.50		655.88	655.88	655.88	655.88
0.57		656.07	655.88	655.88	655.88
0.64		655.88	656.07	655.88	656.07

Table 2: Raw data of highest amplitude harmonic frequencies at each plucking length

ABSOLUTE UNCERTAINTIES

- The absolute uncertainty in the picking length x_1 is $\Delta x_1 = \pm 0.005\text{m}$ as the tape measure has a precision of 1 cm.
- The absolute uncertainty in the frequency f is

$$\Delta f = \pm n \times 0.05 f_1 = \pm n \times 16.48\text{Hz}$$
 (4)

as the frequency search window in the FFT is 5% of the expected fundamental frequency $f_1 = 329.63\text{Hz}$. This uncertainty is multiplied by n because it propagates when the non-fundamental harmonic is calculated by multiplying the fundamental frequency by the harmonic number n . For instance, plucking length $x_1 = 0.01\text{m}$ results in the 2nd harmonic with the highest amplitude, so regardless of the FFT-found fundamental frequency, the absolute uncertainty is $\Delta f = \pm 2 \times 16.48 = \pm 32.96\text{Hz}$.

¹Outlier removed for data processing

3.3 Processed Data

Plucking length x_1 (m)	Average frequency with highest amplitude $\bar{f}$ (Hz)	Uncertainty in $\bar{f}$ $\Delta \bar{f}$ (Hz)
$\Delta x_1 = \pm 0.005\text{m}$	$\bar{f}$ (Hz)	$\Delta \bar{f}$ (Hz)
0.01	655.85	32.96
0.08	655.85	32.96
0.15	655.92	32.96
0.22	655.99	32.96
0.29	983.97	49.44
0.36	655.93	32.96
0.43	655.92	32.96
0.50	655.88	32.96
0.57	655.92	32.96
0.64	655.99	32.96

Table 3: Average frequency with highest amplitude and uncertainties at each plucking length ^{the averages and the uncertainties} ^{FX}

CALCULATIONS

- The average frequency with the highest amplitude $\bar{f}$ was calculated by averaging the frequencies with the highest amplitude across trials, for each variation of the picking length x_1 . For example, for $x_1 = 0.01\text{m}$, the average frequency $\bar{f}$ is

$$\bar{f} = \frac{1}{5} \sum_{n=0.01, \text{trial } n}^5 (f_{n=0.01, \text{trial } n})$$
$$= \frac{1}{5} (655.70 + 655.88 + 655.88 + 655.88 + 655.88)$$
$$= 655.85 \text{ Hz}$$
 (5)

- The uncertainties are due to systematic error from the FFT process rather than random error between trials, therefore the absolute uncertainties are used as in Table 2 as opposed to $\Delta f = \pm 32.96\text{Hz}$. For instance, the absolute uncertainty for picking length $x_1 = 0.01\text{m}$ is 32.96Hz , so $\Delta \bar{f} = 32.96\text{Hz}$.

3.4 Graphical Analysis



Figure 4: Graph of picking length x_1 against average frequency $\bar{f}$

Graphing the processed data, there is a clear peak at the data point (0.29, 983.97), while every other data point has a y-value around $655\text{Hz} \pm 1\text{Hz}$. The error bars are quite large, around $\pm 30\text{--}50\text{Hz}$, however, this does not impair the clear peak at 0.29m.

The graph cannot be linearized within the scope of the IA due to requiring Taylor series approximations (Ratkovsky & Dolby, 1975). Therefore, the data will be compared to the theoretical simulation data, to which a Gaussian function is fitted.

3.5 Comparison with Theoretical Data

Using the method outlined in Section 1.1, the theoretical harmonic frequency with the highest amplitude $f_{\text{max,theor}}$ was calculated for each variation of the picking length x_1 . The collected data was then compared against the theoretical data, as shown in Table 4 and Figure 5.